

Definitions

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1

Definition 1.1 (No-lonely-color graphs). A graph G is no-lonely-colored if there exists an edge-coloring such that

- Every vertex's edges can only have at most two of the same color.
- Every cycle contains no color exactly once.

Definition 1.2 (Cayley Graphs). Let G be a group, and let $S \subseteq G$, construct the graph $C(G, S)$ as follows:

- Every $g \in G$ is a vertex,
- For every pair $s \cdot g_1 = g_2$, where $s \in S$ and $g_1, g_2 \in G$, set an edge between g_1 and g_2 , disallowing identical edges.

$C(G, S)$ is then called a Cayley Graph.

Definition 1.3 (Minimal Cayley Graphs). A Cayley Graph $C(G, S)$ is called minimal when S is the smallest possible subset still generating the group G .

Definition 1.4 (Girth). The Girth of a graph is the length of its shortest cycle, if no cycles exist the girth is set to be ∞ .

Definition 1.5 (Multigraphs). A Multigraph is a graph where it is allowed for two vertices to have multiple edges connecting each other.

Definition 1.6 (Bridges). Let M be a Multigraph and e be an edge of that graph. If removing e from M results in a graph that has more connected components, we call e a bridge.

Definition 1.7 (Hypergraphs). Let V be a set of elements and let $E \subset P(X)$. We call the collection (V, E) , an (undirected) Hypergraph. The subset E is the collection of hyperedges of the Hypergraph.

Definition 1.8 (Chromatic Number). The chromatic number of a (hyper)graph G is the smallest possible number of colors necessary to color the vertices G if adjacent vertices can't have the same color.

Definition 1.9 (r-uniform Hypergraphs). Let (V, E) be a Hypergraph. If each Hyperedge of the Hypergraph contains exactly k elements, we call the Hypergraph k -uniform.

Definition 1.10 (Tranquil Hypergraphs). An r -uniform hypergraph is tranquil if after labeling all vertices (this must be possible) in each hyperedge with the same r labels, every closed walk in the hypergraph, when projected as a multigraph from label to label, is bridgeless.

Geassumeerde voorkennis:

- Grafentheorie
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